

- 4 A laser produces a narrow beam of coherent light of wavelength 632 nm. The beam is incident normally on a diffraction grating, as shown in Fig. 4.1.

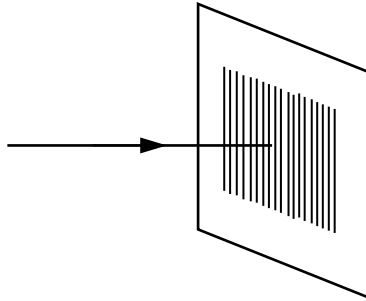


Fig. 4.1

- (a) Describe how diffraction of light takes place at the grating.

The grating consists of many slits/openings and diffraction take place when the light
incident on the grating spread as they pass through each slit/opening in the grating.

[1]

- (b) The diffraction pattern on the screen is shown in Fig. 4.2. The brightest spot is O. The two bright spots closest to O is 3.5 cm away from O.

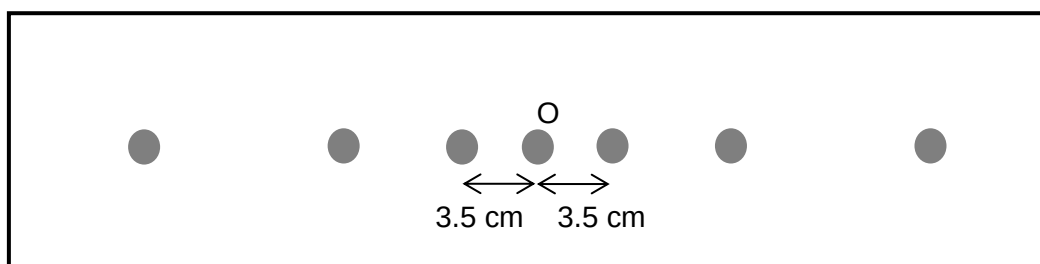


Fig. 4.2 (Not to scale)

The diffraction grating is placed 10 cm from the screen.

Determine the number of lines per metre on the grating.

$$\tan \theta = \frac{3.5}{10} \quad [\text{M1}]$$

$$d \sin \theta = (1)\lambda \quad [\text{M1}]$$

$$\text{no of lines per metre} = \frac{1}{d} = 5.23 \times 10^5 \quad [\text{A1}]$$

number of lines per metre = [3]

- (c) A second laser is directed normally to another diffraction grating with the same number of lines as in (b).

Describe and explain how the new appearance of the diffraction grating pattern will allow the following to be deduced.

- (i) the wavelength of the second laser

If the distance the first order bright spots and the brightest spot at O is increased, the angle of diffraction will increase. Hence the wavelength of the second laser will be longer compared to the first laser.

..... [2]

- (ii) the orientation of the diffraction grating.

The direction of spreading of the bright spots is always perpendicular to the orientation of the slits in the grating. If the bright spots formed on the screen is spread vertically instead of horizontally, this implies that the orientation of the slits for the diffraction grating is rotated by 90 degrees as the slits are now horizontal instead of vertical.

- (d) The diffraction grating in (c) is added directly in front of the first grating such that the orientation of the two diffraction gratings are perpendicular to each other. The diffraction pattern in Fig. 4.3 is observed.

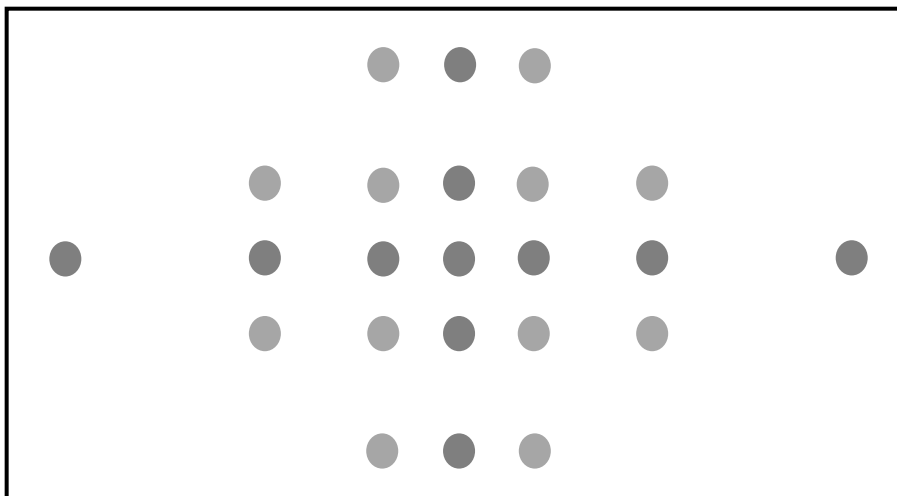


Fig. 4.3

Suggest how the pattern in Fig. 4.3 is formed.

When the light is incident on the first grating, the pattern shown in Fig. 4.2 will be seen.

Each bright spots then acts as source of light for the second grating which results in the pattern formed in Fig. 4.3. The intensity of the bright spots at the corners is much lower compared to the middle portion as they are produced by diffraction of the higher order

[2]

bright spots in Fig. 4.2 or they are spread twice by each grating .

- (e) A student sets up the apparatus in Fig. 4.1 but rotates the diffraction grating by 45° such the laser is no longer normal to the grating.

Suggest and explain whether the position of the brightest spot O in Fig. 4.2 will change.

The position of the bright spot O will not change as the path difference between light coming from adjacent slits in the grating remains zero.

[1]

[Total: 10]

Figure 5: Two stars A and B are separated by a distance of 1.2×10^{10} m as shown in Fig. 5.1. γ is the